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Effect of force-velocity profile-based individualized vs. non-individualized strength training on force-velocity profiles and athletic performance: a systematic review and meta-analysis

Gaojun Li[1,2†], Yang Shen[1,2†], Junlei Lin[1,2] and Wei Li[1,2*]

Abstract

Purpose We aimed to conduct a systematic review with meta-analysis of the effects of force-velocity (FV) profile-based individualized strength training vs. non-individualized strength training on mechanical variables of FV profile and athletic performance.

Methods A computerized systematic literature search was performed in three electronic databases. Studies were included if they analyzed effects of FV profile-based individualized strength training on the metrics of FV measures (theoretical maximal force (F_0), theoretical maximal velocity (V_0), force-velocity imbalance (FV_{imb}) and theoretical maximal power (P_{max})) and athletic performance in adults and tested at least one variable of interest. The effect sizes were determined using a robust variance estimation random-effects model and were reported as Hedge's *g*.

Results Overall, 6 studies were eligible to be included in this meta-analysis. Individualized FV strength training lasted between 7 and 10 weeks, with 2 weekly sessions and 8–28 sets per week. Strength training intensities ranged between 70 and 85% of the one-repetition maximum (1-RM) x 3–10 repetitions resistance training (training load ratio range: 70–100% : 30 − 0%) and velocity-oriented training involved 0–60% body mass x 3–10 repetitions plyometric training (training load ratio range: 70–100% : 30 − 0%). The meta-analysis showed significant and large differences between both training conditions for F_0 ($g = 0.99$, $p < 0.01$), V_0 ($g = 0.78$, $p < 0.01$), FV_{imb} ($g = 0.60$, $p < 0.01$) and vertical jump height ($g = 0.82$, $p < 0.01$) favoring individualized strength training, whereas there was a non-significant effect on P_{max} ($g = -0.12$, $p = 0.42$) and sprint performance ($g = 0.18$, $p = 0.45$),

Conclusion Our findings suggest that FV profile–based individualized training may confer advantages for FV profile optimization and vertical jump performance, the high degree of heterogeneity underscores the need for cautious

†Gaojun Li and Yang Shen contributed equally to this work and share the first authorship.

*Correspondence:
Wei Li
liweiwin@sina.com

Full list of author information is available at the end of the article

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interpretation and further well-controlled studies. Therefore, practitioners should specifically and carefully consider individual FV profile when programming individualized strength training.

Keywords Maximal power, Vertical jump, Imbalance, Maximal force, Maximal velocity

Background

Maximal power output represents the highest rate of mechanical work (force × velocity) that skeletal muscle can generate [1, 2]. By integrating force production and contraction velocity, maximal power output reflects maximal mechanical capability, as it indicates how effectively force can be expressed at high movement speeds during ballistic actions. Theoretically, achieving maximal power output requires an optimal balance between force and velocity, whereas a deficiency in either component compromises power production [3, 4]. Consequently, individual differences in force and velocity development should be considered when designing strength training programs aimed at maximizing power output [3], an approach referred to as force–velocity (FV) profile–based individualized strength training. Within this framework, athletes with force deficits are recommended to prioritize maximal strength development, whereas those with velocity deficits should emphasize training strategies that enhance movement velocity. In contrast, non-individualized strength training applies uniform programs with identical loads and progressions, assuming similar adaptive responses across individuals and potentially leading to under- or excessive stimulation due to unaccounted mechanical differences. Accordingly, FV profile–based individualized strength training has been proposed as a more targeted strategy that aligns training stimuli with each athlete's mechanical profile to optimize maximal power output.

Individual differences in force and velocity development can be quantified using the FV profile. The nonlinear inverse FV relationship was first demonstrated in isolated muscle [5], and the mechanical characteristics of multi-joint movements are likewise governed by intrinsic skeletal muscle properties [6, 7], indicating that the fundamental force–velocity relationship is preserved across single- and multi-joint movement patterns [8, 9]. Samozino et al. [10] introduced practical field-based methods to determine the force–velocity–power profile from a series of loaded vertical jumps, enabling the assessment of key individual mechanical outputs using widely available equipment [11, 12]. Although vertical jumping is the most commonly used assessment modality, FV profile can also be derived from sprint performance by analyzing split times and applying biomechanical modeling to estimate horizontal force and velocity outputs [13]. From the individual FV profile, theoretical maximal force (F_0) represents the force an athlete could generate at zero movement velocity, whereas theoretical maximal velocity (V_0) represents the velocity an athlete could achieve under zero external load. The imbalance between force and velocity capabilities (FV_{imb}) can be quantified by comparing the relative difference between the optimal and actual FV profiles derived from loaded vertical jump testing [3, 4].

Based on FV_{imb}, individuals are classified as force-deficit ($FV_{imb} < 100\%$), well-balanced ($90 < FV_{imb} < 110$), or velocity-deficit ($FV_{imb} > 100\%$), where values below and above 100% indicate insufficient force or velocity expression relative to the theoretical optimal FV profile, respectively [14]. Specifically, force- and velocity-deficit profiles are further subdivided into low and high deficits (force: low deficit: 60–90% and high deficit: < 60%; velocity: low deficit:110–140% and high deficit: > 140%, respectively) [14]. Theoretically, force-deficient individuals are expected to benefit from force-oriented training, whereas velocity-deficient individuals may respond more favorably to velocity-oriented training [14, 15]. In practice, this individualized prescription is typically operationalized through a specific force- to velocity-oriented training load ratio, which defines the relative proportion of training volume allocated to maximal strength versus high-velocity exercises, most commonly expressed as the number of sets prescribed for each exercise category [14]. Most empirical studies have adopted a graded approach based on deficit magnitude, commonly applying ratios of approximately 80%:20% (force: velocity) for high deficits and 70%:30% for low deficits [14, 16, 17]. However, some studies have employed uniform, predominantly force-oriented ratios (e.g., 100%:0% or 80%:20%) regardless of deficit magnitude, highlighting variability in implementation across the literature [18].

A substantial body of evidence indicates that force–velocity (FV) profile–based individualized strength training effectively improves FV imbalance (FVimb) and vertical jump performance [3, 4, 19]. Compared with non-individualized training, this approach has been shown to elicit greater improvements in well-trained and elite athletes [14, 16, 20] as well as in the general population [17]. However, findings from Lindberg et al. [18] reported comparable improvements in vertical jump height following both FV profile–based individualized and non-individualized strength training, indicating inconsistency within the existing literature. Such conflicting findings may be attributed to differences in participants' training background, exercise selection, and intervention design. Addressing these inconsistencies is critical both for advancing theoretical understanding

of force–velocity–specific training adaptations and for guiding evidence-based programming decisions in applied strength and conditioning practice.

Given the growing number of studies on this topic, a contemporary synthesis is warranted to clarify the effects of FV profile–based individualized strength training on both mechanical characteristics of the FV profile, including theoretical maximal force (F_0), theoretical maximal velocity (V_0), FV_{imb}, and athletic performance outcomes such as sprinting and jumping. Therefore, the purpose of this systematic review and meta-analysis was to investigate the effects of FV profile–based individualized strength training on FV profile–related mechanical variables and athletic performance.

Methods

Literature search and selection criteria

This systematic review and meta-analysis was conducted following the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) guidelines [21]. This review was registered in PROSPERO (CRD420251041177).

A systematic search for original studies was performed using PubMed, SPORTDiscus, and Web of Science to identify relevant articles examining the effects of FV profile-based individualized strength training vs. non-individualized strength training on mechanical variables of FV profile and athletic performance. The final search was performed on 12 April 2025 and was carried out independently by two authors (GJ and SY). The search strategy combined terms using Boolean operators (AND, OR) as follows: ("youth" OR "adult" OR "young" OR "teen") AND ("individualized training" OR "resistance training" OR "strength training" OR "weightlifting training" OR "plyometric training" OR "weight training") AND ("jump" OR "hop" OR "sprint" OR "ballistic" OR "force-velocity" OR "force-velocity-power" OR "mechanical properties" OR "mechanical variables" OR "muscular strength" OR "muscle strength" OR "change of direction" OR "COD" OR "CoD" OR "1 RM" OR "strength outcome" OR "strength performance"). Further records were identified through checking the reference lists of included studies.

Table 1 Selection criteria according to PICOS criteria (population, intervention, comparator, outcome, study design

Category	Inclusion criteria	Exclusion criteria
Population	Healthy adults aged 18–65 yrs	Younger and older individuals age < 18 year and > 65 year unhealthy individuals
Intervention	Vertical force-velocity profile-based individualized training was applied, e.g., force-deficit players performed force-oriented training and velocity-deficit players performed velocity-oriented training.	Non-individualized strength training and the duration of intervention < 6 weeks
Comparator	Active (i.e., non-individualized strength training) control	No control group
Outcome	At least one mechanical outcomes (i.e., F_0, V_0, FV_{imb} and P_{max}) of FV profile or athletic performance outcomes (maximal strength, sprint, and vertical jump) was reported.	Failed to report the pre-post means or change scores and their standard deviations numerically or graphically; studies for which authors did not reply to our inquiries sent by email
Study design	Randomized and non-randomized controlled trials	Cross-over design

According to the PICOS framework, eligibility criteria were defined in terms of the population, intervention, comparators, outcomes, and study design (Table 1) [22]. For the population category, studies were eligible if they included healthy adults aged 18–65 yrs. For the intervention, participants performed FV profile-based individualized strength training recommendations (i.e., force-deficit players performed force-oriented training and velocity-deficit players performed velocity-oriented training). Individualized strength training was carried out in terms of FV_{imb} metric derived from an individual's vertical jump FV profile, including countermovement jump (CMJ) or squat jump (SJ). Only studies including an active control group performing non-individualized strength training were considered. At least one mechanical outcome (i.e., F_0, V_0, FV_{imb} and P_{max}) of FV profile or athletic performance outcomes (maximal strength, sprint, and vertical jump) was reported. We considered randomized controlled trials and non-randomized controlled trials. Studies were excluded if they met any of the following criteria: (1) participants were unhealthy or younger than 18 year or older than 65 year; (2) the duration of intervention was less than 6 weeks; (3) no control group was included; (4) failed to report the pre-post means or change scores and their standard deviations numerically or graphically; studies for which authors did not reply to our inquiries sent by email; (5) the study employed a cross-over design, as potential cross-education or/and carry-over effects may influence the outcomes of long-term training interventions.

Duplicate records were removed using the reference management software (Endnote X9.3.2; Stanford University, Stanford, CA, USA). The remaining articles were then independently examined by two authors (GJ and SY). Titles and abstracts were initially assessed according to the predefined eligibility criteria, followed by full-text screening of potentially eligible studies. In terms of any disparities, a third author (LW) was involved in decision making.

Data extraction

All data were extracted by one author (GJ) and independently verified by a second author (SY) using a customized Excel spreadsheet. The following information was extracted: study information (e.g., author(s), year), participant characteristics (e.g., sample size, height, body mass, age, sex, training background), FV profile assessment methods (e.g., countermovement jump (CMJ), squat jump (SJ)), intervention characteristics (e.g., training load ratio, intensity, number of sets and repetitions, intervention duration, and training frequency), outcomes (e.g., maximal strength, sprint velocity or time, jump height or distance, F_0, P_{max}, V_0 and FV_{imb}). When data were unavailable or presented only in graphical form, the corresponding authors were contacted to obtain the required information.

Study quality assessment

A modified version of the Downs and Black Quality Index tool was used to evaluate study quality [23]. This tool has been widely used in sport science research [24, 25] and allows for the evaluation of randomized controlled trials and non-randomized studies [23, 26]. This tool has shown high test-retest reliability (r = 0.88), good internal consistency (k = 0.89), and acceptable inter-rater reliability (r = 0.75) [23]. There are 10 items in the current methodological scale (Table 1), with a scale of 0–1 (0, clearly no; 1, clearly yes; ? unsure) for each item. Each study scored from 0 to 10, with a higher score representing the higher methodological quality of the study. Study quality assessment was completed independently by two researchers (GJ and SY) and potential disagreement was resolved by a third author (LW).

Statistical analyses

R version 4.2.3 (R Core Team, 2023) with meta for (version 4.2.0) was used to carry out the meta-analysis. As indicated in the Cochrane Handbook [27], meta-analysis can be undertaken with a minimum of two studies [28]; however, analyses were conducted only when at least three studies were available [29]. For each FV profile variable (F_0, V_0, P_{max} and FV_{imb}) and athletic performance outcome (vertical jump, sprint, and CoD), differences between individualized strength training and non-individualized strength training were examined using standardized effect sizes (ES), with Hedges' g adjustment for small sample bias [30]. ES is presented as Hedges' g ES and 95% confidence interval (CI). Individualized strength training was intended to increase the indicators of F_0 and FV_{imb} and decrease the indicator of V_0 in force-deficit players, whereas it was intended to decrease the indicators of V_0 and FVi_{mb} and increase the indicator of F_0 in velocity-deficit players. Thus, pre- and post-intervention data for force-deficit, velocity-deficit, and control group were extracted. To ensure consistency in interpretation, effect sizes for F_o, V_o, and FV_{imb} were standardized so that positive values reflected favorable adaptations across all variables. Due to the limited number of studies examining sprint performance, a meta-analysis was not performed for this outcome. When studies included multiple intervention groups, the sample size of the control group was divided to enable appropriate comparisons with each intervention group [27]. The magnitude of the ESs was interpreted according to the following scale: trivial (< 0.20), small (0.20–0.49), moderate (0.5–0.79), large (> 0.80) [31].

The exact number of subgroups can only be determined once all studies meeting the inclusion criteria have been identified, as subgroup formation depends on the actual distribution of study characteristics within the included studies. Without access to the final dataset, it is difficult to predetermine factors such as intervention duration, or study population. Identifying subgroups after study selection ensures analyses accurately reflect the heterogeneity of the evidence, rather than imposing arbitrary or unsubstantiated subgroup structures. Heterogeneity was evaluated using I^2 statistic, with I^2 values of 0%, 25%, 50%, and 75% indicating no, low, moderate, and high heterogeneity, respectively [32]. Due to the limited number of included studies, a quantitative analysis (e.g., meta-regression) of training-configuration variables (such as training load ratio, training frequency, intensity, and intervention duration) was not feasible. Therefore, we conducted a systematic review focusing on these variables to explore their potential role in modulating the adaptations to individualized force–velocity profile–based training. Statistical significance was set at $p \leq 0.05$.

Results

Study characteristics

The search process initially identified 8514 studies, of which 246 duplicate records were removed. Following screening titles and abstracts, 36 full-text articles were screened for eligibility. According to the predetermined inclusion criteria, 6 studies were finally included in the meta-analysis (Fig. 1). A total of 259 participants were included across all studies, and all participants of the included studies were healthy adults with a mean age of 21.82 years. Of the 6 included studies, 5 involved elite athletes [14, 16–18, 33] and 1 involved non-athletes [34].

Regarding the training load ratio, two distinct strategies were identified. Strategy 1 applied different force-to-velocity-oriented training load ratios according to the players' force/velocity deficit. Specifically, four studies reported ratios of approximately 80% force : 20% velocity for players with a high force/velocity deficit and 70% force : 30% velocity for players with a low force/velocity deficit [14, 16, 17, 33]. Strategy 2 applied the same

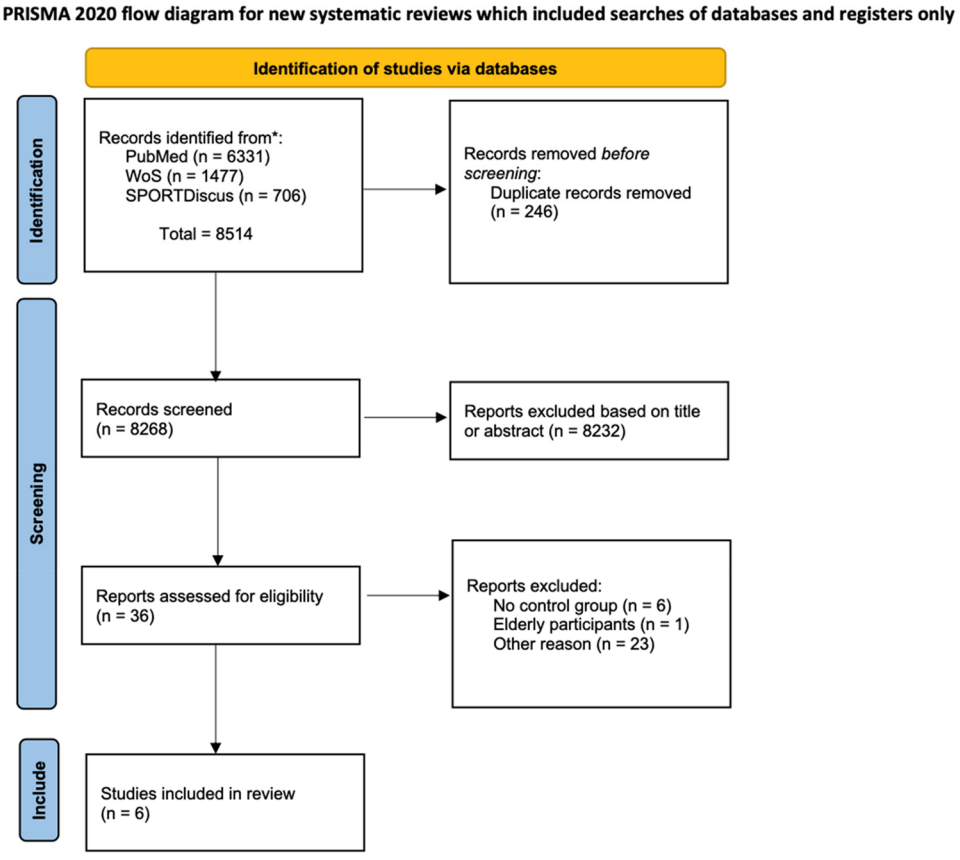


Fig. 1 Flow chart illustrating the different phases of the search and study selection

force- to velocity-oriented training load ratio for players with both high and low force/velocity deficits. In these studies, force-oriented loading ranged from approximately 80% : 20% to 100% : 0%, regardless of deficit classification (Table 2).

Intervention durations ranged from 7 to10 weeks, with a training frequency of 2 sessions per week. In most included studies, participants in the force/velocity-deficit and well-balanced groups performed a total of 18 training sets per week (3 sets x 6 exercises x 3–10 repetitions), which were evenly divided into 2 sessions (3 sets x 3 exercises per session) [14, 16, 17, 33]. In contrast, two studies applied different weekly training volumes, with the number of sets ranging from 8 to 28 across groups [18, 34] (Table 2).

The median score of the modified Downs and Black checklist ranged from 8 to 9 points, with a median score of 9 points (Table 3). All studies were carefully identified as being of good methodology quality.

Meta-analysis results

In terms of F_0, 9 outcomes from 5 studies were used to conduct the meta-analysis. Figure 2 exhibits a large and statistically significant effect in favor of individualized strength training ($g = 0.99$, 95% CI [0.48, 1.50], $p < 0.01$), with a high heterogeneity ($I^2 = 59.09\%$).

In terms of V_0, the effect of individualized strength training vs. non-individualized strength training was investigated in 5 studies for 9 outcomes. Figure 3 displays a large and statistically significant effect in favor of individualized strength training ($g = 0.78$, 95% CI [0.29, 1.27], $p < 0.01$), with high heterogeneity ($I^2 = 55.23\%$).

In terms of P_{max}, 9 outcomes from 5 studies were used to conduct the meta-analysis. Figure 4 exhibits a trivial and non-significant effect of individualized strength training compared with non-individualized training (g = -0.12, 95% CI [-0.44, 0.20], $p = 0.42$), with low heterogeneity ($I^2 = 12.41\%$).

In terms of FV_{imb}, the effect of individualized strength training vs. non-individualized strength training was investigated in 4 studies for 9 outcomes. Figure 5 displays a large and statistically significant effect in favor of individualized strength training ($g = 0.60$, 95% CI [0.23, 0.98], $p < 0.01$), with moderate heterogeneity ($I^2 = 27.51\%$).

In terms of vertical jump performance, 15 outcomes from 6 studies were used to conduct the meta-analysis. Figure 6 shows a large and statistically significant effect in favor of individualized strength training ($g = 0.82$,

Table 2 The characteristics of included studies

Study	Participants	FV profile test	Training load ratio	Training load	Training length and frequency	Assessment outcomes	Load ratio strategy
Álvarez et al. (2019)	*N* = 46, Female professional ballet dancers, Age: 18.9 ± 1.1 yrs, Mass: 54.8 ± 6.1 kg, Height: 1.64 ± 8.4 m	CMJ	EG-FD: HFD: 83%(FOX) : 17% (VOX), LFD: 67% (FOX) : 33% (VOX) CG: no training	EG-FD: Total sets: 18 sets/ week, FOX: 3 s x 6 rep RT : 80–90% 1RM PT: 80–85% BM VOX: 3 s x 6 rep PT: 0–10% BM CG: no training	9 weeks, 2 essions/week	CMJ, FV_{imb} $V_{0,}$ $F_{0,}$ P_{max}	Strategy 1
Barrera-Domínguez et al. (2023)	*N* = 30, Male national level basketball players, Age: 22.8 ± 5.73 yrs, Mass: 86.25 ± 14.5 kg, Height: 1.86 ± 0.07 m	CMJ	EG-FD: HFD: 67%(FOX) : 33% (VOX), LFD: 67% (FOX) : 33% (VOX) EG-VD: HVD: 17%(FOX) : 83% (VOX), LVD: 17%(FOX) : 83% (VOX), CG: 50%(FOX) : 50% (VOX),	EG: Total sets: 18 sets/ week FD: FOX: RT : 3 s x 5–10%VL x 70–95% 1RM PT: 3 s x 3 reps x 80% BM VOX: PT: 3 s x 4 reps x 0–10% BM VD: FOX: RT: 3 s x 3 reps x 50–65% 1RM VOX: PT: 3 s x 3–5 reps x BM CG: Total sets: 18 sets/ week FOX: RT: 3 s x 10% VL x 70–90% 1RM VOX: PT: 3 s x 4–5 reps x 0–10%BM	8 weeks, 2 seeisons/week	CMJ, Single-leg DJ , Triple hop, 10 m sprint, FV_{imb} , $V_{0,}$ $F_{0,}$ P_{max}	Strategy 2
Jiménez-Reyes et al. (2017)	*N* = 84, Semi-professional soccer and rugby players, Age: 23.1 ± 4.4 yrs, Mass: 75.5 ± 8.5 kg Height: 1.79 ± 0.05 m	SJ	EG-FD: HFD: 83%(FOX) : 17% (VOX), LFD: 67% (FOX) : 33% (VOX) EG-VD: HVD: 17%(FOX) : 83% (VOX), LVD: 33%(FOX) : 67% (VOX), CG: 60%(FOX) : 40% (VOX),	EG: Total sets: 18 sets / week, FD: FOX: RT : 3 s x 80–95% 1RM PT: 3 s x 70–80% BM VOX: PT: 3 s x 0–45% BM VD: FOX: RT: 3 s x 65% 1RM VOX: PT: 3 s x 3–5 reps x 0–45%BM CG: Total sets: 18 sets / week, RT : 3 s x 80–95% 1RM PT: 3 s x 70–80% BM VOX: PT: 3 s x 0–10% BM	9 weeks, 2 seeisons/week	SJ, FV_{imb} , $F_{0,}$ $V_{0,}$ P_{max}	Strategy 1
Lindberg et al., (2021)	*N* = 40, Male national level team sport players, Age: 20.0 ± 4.0 yrs, Mass: 83.0 ± 13.0 kg Height: 1.84 ± 0.09 m	SJ	EG-FD: 83%(FOX) : 17% (VOX), EG-VD: 21% (FOX) : 79% (VOX) CG: 46%(FOX) : 54% (VOX),	Total sets: 18 sets / week EG-FD: FOX: 3–10 reps x 1–6 RIR VOX: 5 reps x 50–70% 1RM EG-VD: 28 sets/week FOX: 3–8 reps x 1–2 RIR VOX: 5–10 reps x negative-50% 1RM CG: Total sets: 28 sets / week, FOX: 3–10 reps x 1–6 RIR VOX: 5–10 reps x negative-50% 1RM	8 weeks, 2 sessions/week	SJ, CMJ FV_{imb}, F0, V0, $P_{max,}$ 1RM Squat , 10 m sprint 30 m sprint	Strategy 2

Table 2 (continued)

Study	Participants	FV profile test	Training load ratio	Training load	Training length and frequency	Assessment outcomes	Load ratio strategy
Simpson et al., (2021)	*N* = 29, Professional rugby league players, Age: 24.00 ± 3.00 yrs, Mass: 94.90 ± 21.60 kg, Height: 1.81 ± 0.60 cm	SJ	EG-FD: HFD: 83%(FOX) : 17% (VOX), LFD: 67% (FOX) : 33% (VOX) EG-VD: HVD: 17%(FOX) : 83% (VOX), LVD: 33%(FOX) : 67% (VOX), CG: 60%(FOX) : 40% (VOX),	EG: Total sets: 18 sets / week, FD: FOX: RT : 3s x ≥ 80%1RM PT: 3s x 65–70% BM VOX: PT: 3 s x 0–10% BM VD: FOX: RT: 3 s x 65% 1RM VOX: PT: 3 s x 3–5 reps x 0–45%BM CG: Total sets: 18 sets / week, RT : 3 s x ≥ 80% 1RM PT: 3 s x 30%-65% BM VOX: PT: 3 s x 0–10% BM	8 weeks, 2 sessions/week	SJ, FV_{imb}, F0, V0, P_{max}, 3RM Squat, 10 m sprint 20 m sprint	Strategy 1
Zubčić & Vučetić (2022)	*N* = 30, Male university students, Age: 21.97 ± 2.25 yrs, Mass: 78.67 ± 9.11 kg, Height: 1.80 ± 0.07 cm	SJ	EG-FD: 100%(FOX) : 0% (VOX), CG: 0%(FOX) : 100% (VOX),	EG-FD: Total sets: 8–10 sets / week, FOX: 4–5 s x 5 reps x 80–85% 1RM CG: Total sets: 14–18 sets / week, 7–9 s x 6 reps x negative 25% BM	7 weeks, 2 session/week	SJ, FV_{imb}, F0, V0, P_{max}	Strategy 2

FV Force-velocity, *N* Number, *CMJ* Countermovement jump, *FD* Force-deficit, *VD* Velocity-deficit, *HFD* High force deficit, *LFD* Low force deficit, *FOX* Force-oriented exercise, *VOX* Velocity-oriented exercise, *S* Sets, *Reps* Repetitions, *RM* Repetition maximal, *PT* Plyometric training, *BM* Body mass, *EG* Experiment group, *CG* Control group, FV_{imb} Force-velocity imbalance, V_0 Theoretical maximal velocity, F_0 Theoretical maximal force, P_{max} Theoretical maximal power, *SJ* Squat jump, *RIR* Repetition in reserve

Table 3 Results of methodological quality of the included studies

Study	Question number										Score
	1	2	3	4	5	6	7	8	9	10	
Álvarez et al. (2019)	1	1	1	1	1	1	?	1	1	1	9
Barrera-Domínguez et al. (2023)	1	1	1	1	1	1	?	1	1	1	9
Jiménez-Reyes et al. (2017)	1	1	0	1	1	1	?	1	1	1	8
Lindberg et al., (2021)	1	1	1	1	1	1	?	1	1	1	9
Simpson et al., (2021)	1	1	1	1	1	1	?	1	1	1	9
Zubčić & Vučetić (2022)	1	1	1	1	1	1	?	1	1	1	9

Average score = 8.8, Median score = 9

Item: (1) the objectives of the study were clearly reported, (2) the main outcomes to be assessed were clearly reported, (3) the characteristics of the participants were clearly reported, (4) the main ůndings were clearly reported, (5) the estimates of the random variability in the data for the main outcomes were clearly reported, (6) the actual probability values were clearly reported, (7) can the participants represent the entire population, (8) if any of the results of the study were based on 'data dredging' was this made clear?, (9) were the statistical tests appropriate, (10) were the main outcome measure accurate. 1 the item was clearly reported, 0 the item was not clearly reported, ? unknown

95% CI [0.50, 1.13], $p < 0.01$), with high heterogeneity ($I^2 = 53.52\%$).

In terms of sprint performance, 6 outcomes from 3 studies were used to conduct the meta-analysis. Figure 7 exhibits a trivial and significant effect in favor of individualized strength training ($g = 0.18$, 95% CI [-0.38, 0.74], $p = 0.45$), with moderate heterogeneity ($I^2 = 47.85\%$).

Discussion

This systematic review and meta-analysis is the first to examine the effects of FV profile-based individualized strength training on FV profile-related mechanical variables and athletic performance. Overall, the findings of the present study support the hypothesis that individualized strength training elicits significantly

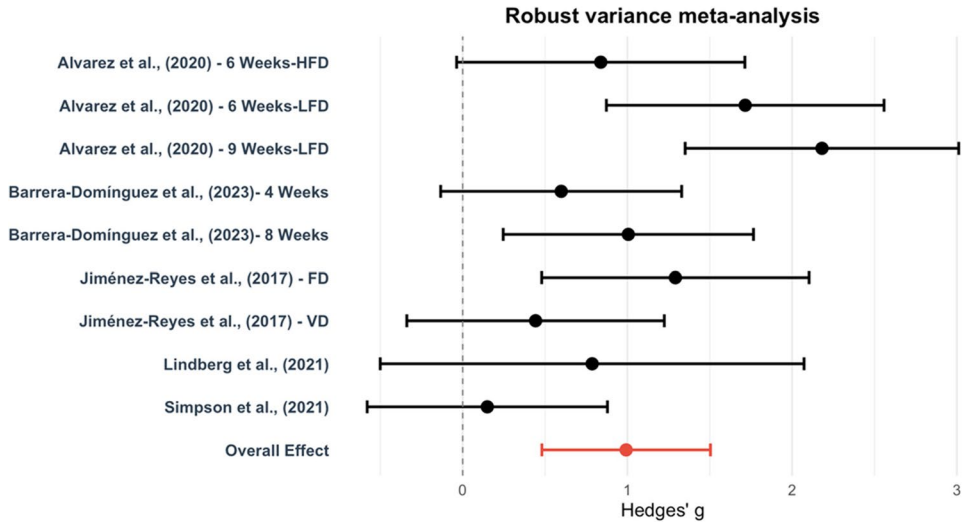


Fig. 2 The forest plot from the meta-analysis of the effects of force-velocity profile-based individualized strength training vs. non individualized strength training on theoretical maximal force. LFD = low force deficit; HFD = high force deficit; FD = force deficit; VD = velocity deficit

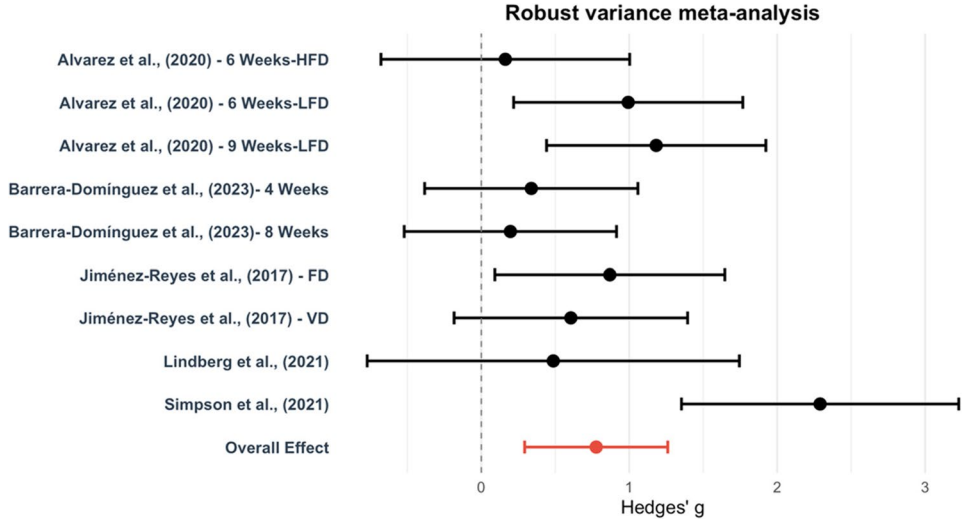


Fig. 3 The forest plot from the meta-analysis of the effects of force-velocity profile-based individualized strength training vs. non individualized strength training on theoretical maximal velocity. LFD = low force deficit; HFD = high force deficit; FD = force deficit; VD = velocity deficit

greater improvements in FV imbalance (FV_{imb}), F_0, V_0, and vertical jump performance compared with non-individualized training. We therefore recommend that FV profile-based individualized strength training should be considered when programming strength training, rather than employing the 'one-size-fits-all' program. From an applied perspective, practitioners should carefully consider key programming variables, including the training load ratio, overall training load, intervention duration, and training frequency. In particular, athletes with greater force- or velocity-related deficits may benefit from higher force- or velocity-oriented training load ratios and longer intervention durations. In summary, while the present meta-analysis suggests that FV profile–based individualized training may confer advantages for FV profile optimization and vertical jump performance, the high degree of heterogeneity underscores the need for cautious interpretation and further well-controlled studies.

Effect on FV profile

Imbalances between force and velocity development are common in sportspersons, as many sports predominantly emphasize either force- or velocity-oriented demands. Our findings confirm that FV profile-based individualized strength training program is a superior strategy for improving variables of FV profiles and optimizing the FV profile. This effectiveness stems from the principle of training specificity; by identifying whether an individual is constrained by a 'FD' or a 'VD', the intervention

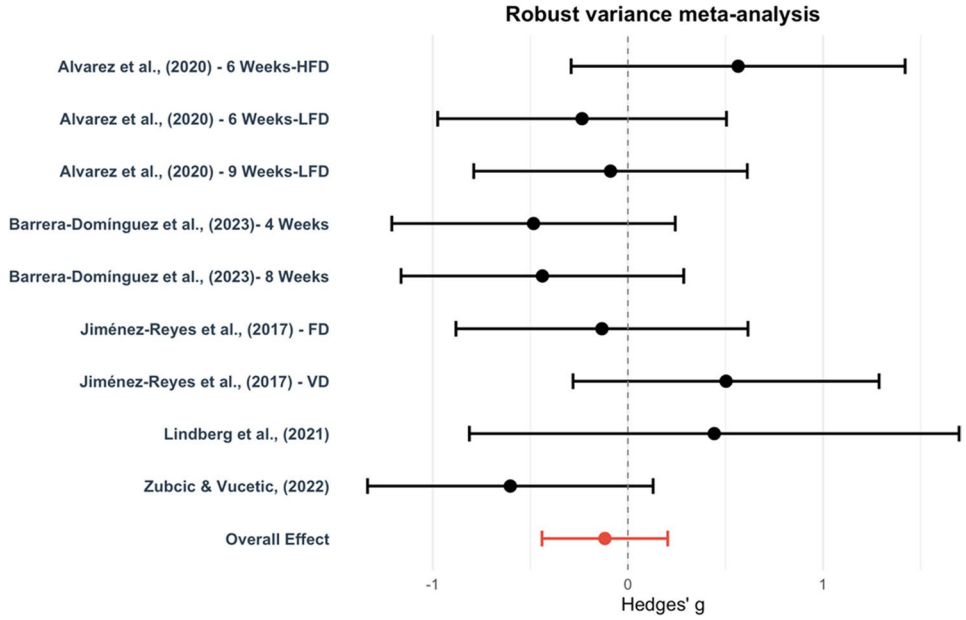


Fig. 4 The forest plot from the meta-analysis of the effects of force-velocity profile-based individualized strength training vs. non individualized strength training on theoretical maximal power. LFD = low force deficit; HFD = high force deficit; FD = force deficit; VD = velocity deficit

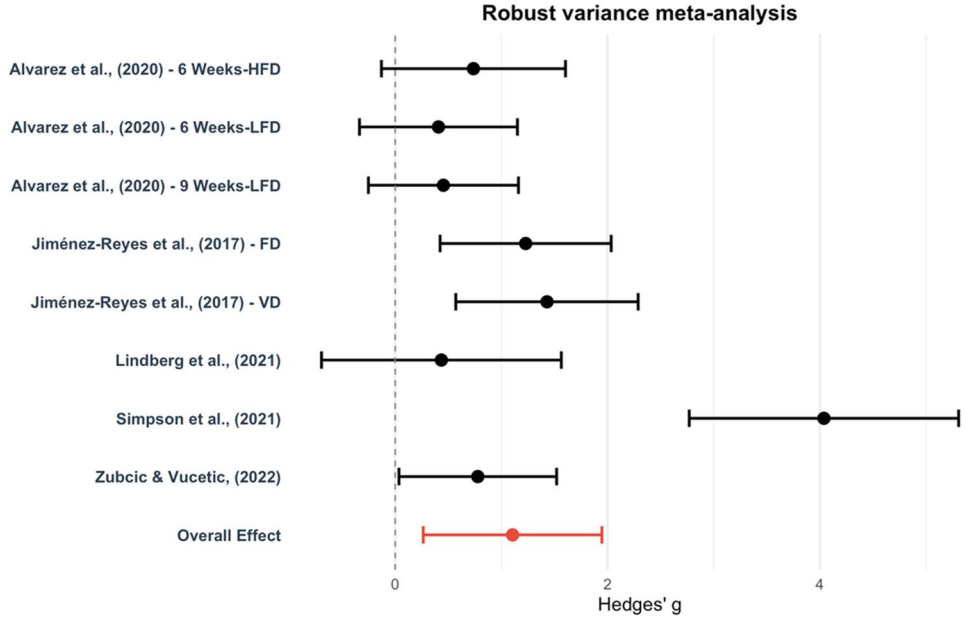


Fig. 5 The forest plot from the meta-analysis of the effects of force-velocity profile-based individualized strength training vs. non individualized strength training on force-velocity imbalance. LFD = low force deficit; HFD = high force deficit; FD = force deficit; VD = velocity deficit

provides a tailored mechanical stimulus that directly addresses their specific physiological limitations. Consequently, this targeted approach facilitates a more rapid convergence of the athlete's actual FV profile toward their theoretical optimum, thereby enhancing mechanical efficiency. This optimization explains why individualized groups exhibited more pronounced reductions in FV_{imb} compared to traditional, non-individualized protocols, where the training stimulus often fails to align with the unique neuromuscular requirements of each participant.

Interestingly, we did not find statistically significant differences in P_{max} following both interventions, a finding that aligns with the theoretical framework of FV profile. Biomechanically, P_{max} reflects the maximal theoretical ceiling of power output, whereas FV_{imb} characterizes the orientation of the force-velocity profile. Our findings indicate that individualized training functions primarily

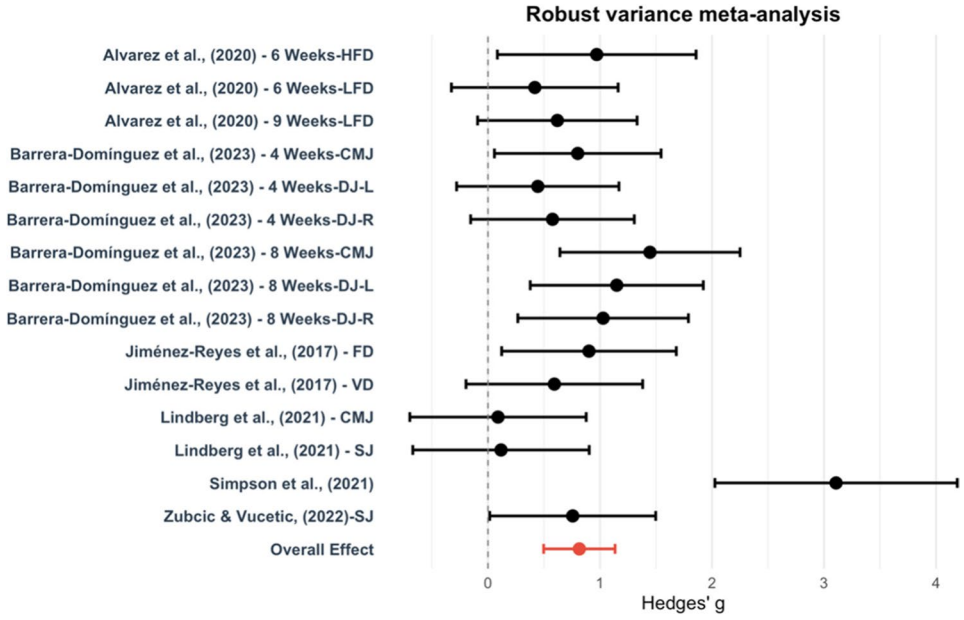


Fig. 6 The forest plot from the meta-analysis of the effects of force-velocity profile-based individualized strength training vs. non individualized strength training on vertical jump. LFD = low force deficit; HFD = high force deficit; FD = force deficit; VD = velocity deficit

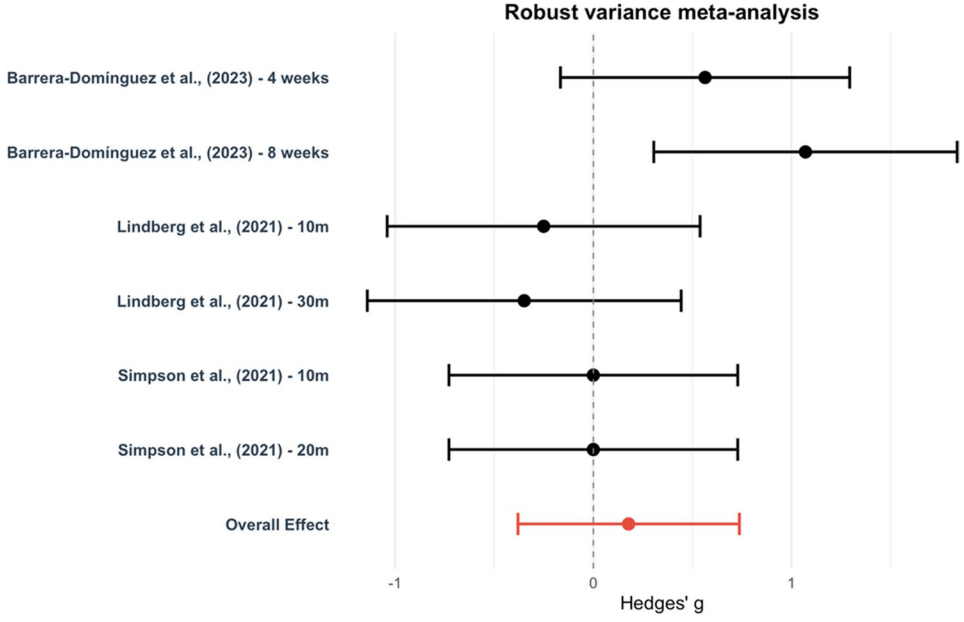


Fig. 7 The forest plot from the meta-analysis of the effects of force-velocity profile-based individualized strength training vs. non individualized strength training on sprint performance. LFD = low force deficit; HFD = high force deficit; FD = force deficit; VD = velocity deficit

as a 'slope-correction' strategy. By specifically targeting an individual's force or velocity deficit, the intervention realigns the profile slope toward its theoretical optimum without necessarily expanding the absolute power envelope. This implies that the initial phase of individualized F-V training favors the qualitative reorganization of force-velocity characteristics over the quantitative augmentation of maximal power production.

Effect on vertical jump

Resistance training has been proven as an effective strategy to enhance vertical jump performance in populations with varied training backgrounds [22, 35, 36]. Thus, practitioners put more attention on improving the effectiveness of resistance training, introducing several novel resistance training strategies including velocity-based training [37], cluster training [38], complex training [39] and so on. Unfortunately, less research focuses

on FV profile-based individualized strength training, leaving coaches and athletes unclear of its effectiveness. We found a significantly greater improvement in vertical jump height in individualized strength training group compared with non-individualized group ($g = 0.82$, $p < 0.01$) (Fig. 6). Notably, the enhancement in vertical jump performance was primarily driven by the significant optimization of FV_{imb}, rather than changes in P_{max}, which showed no significant alterations. This result underscores that vertical jump performance is heavily contingent upon their FV profile balance. From a biomechanical perspective, a shift toward an 'optimal' FV profile allows for a more efficient application of force during the take-off phase, thereby maximizing the conversion of muscular potential into vertical displacement. Consequently, our findings suggest that for many athletes, optimizing FV profile may be just as critical as, if not more so than, increasing raw power output for improving vertical jump performance.

Characteristics of individualized strength training intervention

Training load ratio

A systematic review aimed at identifying the characteristics of effective FV profile-based individualized training programs may provide valuable practical guidance for practitioners. The core principle of FV profile-based individualized training highlights the specific individual demands based on the status of the development of force and velocity. For example, athletes characterized by a force deficit are primarily limited by maximal force production and should therefore emphasize maximal strength–oriented exercises within their training programs. Therefore, determining the appropriate proportion of force- and velocity-oriented exercises-commonly referred to as the training load ratio-is of particular importance. Across all included studies, the training load ratio was operationalized as the proportion of total training volume allocated to force-oriented vs. velocity-oriented exercises, typically quantified by the number of sets performed for each exercise category. For example, when a training session consisted of three sets of back squats at 90% of one-repetition maximum (1RM), three sets of unilateral squats at 90% 1RM, and three sets of squat jumps, the resulting force-to-velocity training load ratio was approximately 70%:30%, reflecting six force-oriented sets and three velocity-oriented sets.

Marked variability in training load ratios was observed across the included studies, ranging from approximately 70%:30% to 100%:0%. Different training load ratios (approximately 70% : 30% and 80% : 20%) was used in three included studies, taking into account the extent of force or velocity deficits [14, 16, 17], whereas the same training load ratios also utilized in both high- and low-force/velocity deficit players [18, 34]. Unfortunately, as several factors (e.g., intensity, training context, etc.) also contribute to training gains, it is unreasonable and blunt to directly compare training gains across studies that utilized different training load strategies. However, individuals with higher velocity or force deficits require a greater effort to shift the FV profile, thus higher training loads may be more effective than lower training loads. We therefore recommend training load ratio of 80% : 20% and 70% : 30% for players with high and low power/velocity deficit, respectively.

Training loading

Physical stress during training is largely mediated by the mechanical load applied, which ultimately determines neural and muscular adaptations, including changes in maximal strength, strength-power, power and speed [40, 41]. The magnitude and nature of this physical stress are governed by the interaction between training intensity and volume [42], highlighting training load as a key determinant of neuromuscular performance adaptations [43]. As mentioned above, resistance training at 80–85% 1RM has been reported as the optimal intensity for enhancing muscle strength in trained adults and athletes [44–47]. Consistent with this evidence, studies that reported significant improvements in FV_{imb} typically prescribed maximal strength exercises at intensities ranging from 80% to 95% 1RM within individualized strength training interventions [14, 16, 17, 34]. In terms of velocity-oriented training, interventions commonly involved unloaded or lightly loaded vertical jumping (0–10% of body mass), whereas power-oriented exercises typically consisted of loaded vertical jumps using external loads corresponding to approximately 10–45% of body mass [14, 16, 17, 20, 34]. In terms of training volume, most studies implemented a total weekly volume of 18 sets, generally organized as three sets per exercise, with two to six repetitions per set across six exercises per week [14, 16, 17]. Applying identical set volumes across exercises facilitates both the calculation of the training load ratio and the practical organization of training programs. It is well documented that maximal strength fluctuates frequently because of fatigue, substantially influencing real-time effort and physical stress [48, 49], and subsequent neuromuscular adaptations. Accordingly, monitoring concentric movement velocity provides immediate feedback that can be used to adjust training loads and regulate effort during training sessions [50]. This autoregulatory method has been implemented within FV profile-based individualized training frameworks and has been associated with positive adaptive outcomes [16]. Consequently, we recommend the incorporation of instantaneous velocity monitoring as a part of individual FV_{imb} training routine

to augment adaptations while simultaneously mitigating excessive fatigue.

Training frequency

Meta-regression evidence by Pelland et al. [51] indicates a modest positive association between weekly training frequency and strength gains, suggesting that higher frequencies may confer additional benefits when training volume is appropriately controlled. Training frequency is influenced by multiple factors, including training intensity, volume, training background, and accumulated fatigue [52]. As training frequency increases, mechanical stimulus is amplified, but fatigue also accumulates, thereby necessitating longer recovery periods. Resistance training performed at least twice per week has been shown to effectively improve maximal strength while allowing sufficient recovery between sessions. In all 6 included studies, the projected training frequency was 2–3 times per week. However, one study reported that actual training frequency was only 1–2 times per week, resulting in non-significant changes in FV_{imb} and vertical jump performance [18]. Although a higher training frequency (> 2–3 times/week) may theoretically facilitate greater neuromuscular adaptations, they may also increase fatigue accumulation and injury risk. To date, no studies have directly compared different training frequencies within FV profile–based individualized strength training interventions. Based on the available evidence, a training frequency of two to three sessions per week appears to be both sufficient and practical for optimizing the balance between force and velocity adaptations.

Training length

Longer training durations are generally associated with greater cumulative mechanical loading and physical stress, thereby facilitating enhanced neuromuscular adaptations [53]. Although short-term resistance training interventions (< 24 weeks) are effective for improving muscle strength, strength gains typically plateau within approximately 9–12 weeks of training [54]. In the present review, all included studies implemented individualized strength training interventions lasting 7–10 weeks, with minimal variation in training duration. While significant improvements in FV_{imb} were observed, these improvements did not necessarily indicate that an optimal FV profile was achieved in all participants.

Notably, individuals with greater force deficit appeared to respond differently to individualized strength training than those with smaller deficits. Previous evidence indicates that players with high force deficits exhibited approximately twice the magnitude of FV_{imb} change compared with those with low force deficits following a 9-week intervention [17]. However, because the discrepancy between the actual and optimal FV profiles is larger in individuals with high force deficits, a longer intervention duration and greater cumulative training exposure may be required to fully correct the imbalance.

Supporting this notion, Barrera-Domínguez et al. [16] found that all 7 players with low-force (n = 2) or low-velocity (n = 5) deficits reached their optimal FV profile following an 8-week intervention, whereas approximately half of the players with high force deficits still exhibited residual imbalance. Similar findings were observed following a 9-week individualized training intervention [14]. Therefore, the optimal training required to achieve optimal F-V profile is largely determined by the initial magnitude of FV_{imb} [20], which should be considered when designing and programming training intervention and further studies are required to investigate the optimal training length for low- and high-force/velocity deficit players.

Limitations

To the best of our knowledge, this is the first meta-analysis to examine the effects of FV profile-based individualized strength training on FV profile-related mechanical variables and vertical jump performance. Nevertheless, several limitations of this review should be acknowledged. First, the limitations of FV profile method have been widely debated [10, 13, 55]. The most critical issue, previously examined, concerns the poor reliability of FV profile, which reduces the ability to identify the level of neuromuscular development. However, this issue was mainly caused by non-standardization in testing procedures, including significant variations in starting position and jump height, unfamiliarity with the jump movement, and other related factors [15]. The noise in force-velocity profile testing can be reduced by averaging multiple trials for each load, increasing the investigated velocity range, standardizing starting positions and expanding the number of experimental conditions to establish the force-velocity relationship [15]. Although the limitations cannot be completely ignored, the FV profile method provides valuable insights into players' force-velocity characteristics and the optimal strategy to enhance ballistic performance. Second, the limited number of eligible studies precluded a precise determination of the optimal training dose required to maximize improvements in FV_{imb}. This represents a relevant limitation, as optimal training load and programming parameters are likely to differ according to the magnitude and direction of individual FV deficit.

Conclusion

The findings of this meta-analysis suggest that significant improvements in FV_{imb} and vertical jump performance were observed following FV profile-based individualized strength training. Therefore, practitioners should

carefully consider individual FV profile when programming strength training. Moreover, practitioners also need to consider the configuration of individualized strength training including training load ratio, training load, training length, and training frequency. It is recommended that a higher training load ratio and longer training length should be performed in players with higher force/velocity deficits. Overall, our results support the practical value of the FV profile strategy in terms of improving optimal FV profile and ballistic performance.

Abbreviations

FV	Force-Velocity
F_0	Theoretical Maximal Force
V_0	Theoretical Maximal Velocity
P_{max}	Theoretical Maximal Power
FV_{imb}	Force-Velocity Imbalance
1-RM	One-Repetition Maximum
CMJ	Countermovement Jump
SJ	Squat Jump
PRISMA	Preferred Reporting Items for Systematic Reviews and Meta-Analyses
PICOS	Population, Intervention, Comparator, Outcomes, Study Design
PROSPERO	International Prospective Register of Systematic Reviews
ES	Effect Size
CI	Confidence Interval
RVE	Robust Variance Estimation
CoD / COD	Change of Direction
LFD	Low Force Deficit
HFD	High Force Deficit
FD	Force Deficit
VD	Velocity Deficit

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Authors' contributions
Gaojun Li was responsible for article search and article review. Yang Shen was responsible for data extraction and using a customized Excel spreadsheet. Junlei Lin was responsible for proposing important amendments to the article. Wei Li was responsible for resolving any differences we had in the article.

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Declarations

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Not applicable.

Consent for publication
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The authors declare no competing interests.

Author details
[1]School of Strength and Conditioning Training, Beijing Sport University, Beijing, China
[2]Force-velocity Laboratory, Beijing Sport University, Beijing, China

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